BIOSENSIA-DC: A DRUGCLIP-BASED COMPUTATIONAL PROTOTYPE FOR TARGET FISHING APPLIED TO BIORECEPTOR DISCOVERY

First Author[1] - e-mail
Second Author[2] - e-mail
Third Author[3] - e-mail
[1]Universidade Federal do Rio Grande – Rio Grande – RS
[2]Universidade Federal do Rio Grande – Santo Antônio da Patrulha – RS
[3]Universidade Federal do Rio Grande – São Lourenço do Sul – RS

Abstract.
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1. INTRODUCTION

1.1 Biosensors and the bioreceptor-selection problem

A biosensor can be understood as an analytical device that couples a biological recognition mechanism to a signal-transduction mechanism. The biological component is responsible for interacting with the analyte of interest, while the transducer converts that interaction into a measurable signal. In this work, we use the terms *bioreceptor* and *biorecognition element* in this general sense. Common examples include enzymes, antibodies, aptamers, receptors, and other biomolecular structures capable of selective molecular recognition (Eggins, 1996; Banica, 2012; Morales and Halpern, 2018). The nature of this recognition element strongly influences the sensitivity, selectivity, reproducibility, and practical applicability of the resulting biosensor.

Selecting a suitable bioreceptor is therefore one of the central design decisions in biosensor development. This decision is not merely a matter of choosing a molecule that can bind the analyte: the candidate must also be compatible with immobilization, signal generation, stability requirements, sample conditions, and the intended sensing platform. Enzyme-based biosensors, immunosensors, and aptamer-based sensors offer different advantages and limitations, but each case still requires the identification of a recognition mechanism that is selective enough for the target application (Marques and Yamanaka, 2008; Morales and Halpern, 2018).

The broader BioSensIA project addresses this problem as a computational discovery task. Given an analyte molecule, the goal is to identify proteins, protein domains, binding pockets, or other biomolecular structures that may interact with it and therefore deserve further investigation as candidate bioreceptors. In this formulation, the computational system is not expected to replace experimental validation. Rather, it is intended to reduce the search space:

instead of manually inspecting many unrelated biological databases and structural records, the user receives a prioritized list of candidates that can be examined with additional structural, biochemical, and bibliographic evidence. This ranking-oriented view is particularly useful when the biological interactions relevant to sensing are not already obvious from the literature.

BioSensIA-DC is one computational component of this broader workflow. It focuses on the structural target-fishing step: given a query molecule, it ranks candidate protein pockets according to a learned compatibility score. The prototype is based on DrugCLIP, a contrastive representation-learning model originally proposed for virtual screening, and on Uni-Mol-style three-dimensional molecular and pocket representations (Gao et al., 2023; Zhou et al., 2023). BioSensIA-DC uses DrugCLIP as a representation-learning foundation, but extends it with a new target-fishing workflow, data organization strategy, and retrieval implementation. In the original virtual-screening direction, a protein pocket is used as a query to rank molecules. BioSensIA-DC uses the same molecule–pocket compatibility idea in the reverse direction: the molecule becomes the query, and protein pockets become the candidates to be ranked. The resulting ranked list is intended to support bioreceptor discovery, not to provide a definitive proof of binding or biosensor performance.

1.2 Virtual screening, target fishing, and representation learning

Virtual screening and target fishing are closely related computational tasks, but they differ in the direction of the query. In structure-based virtual screening, the starting point is usually a biological target, often represented by a protein binding pocket, and the goal is to rank molecules from a candidate library according to their predicted compatibility with that target. In target fishing, the direction is reversed: the starting point is a molecule, and the goal is to identify proteins, protein domains, binding sites, or other biological targets that may interact with it. In both cases, the practical output is a ranked list, but the biological interpretation is different. Virtual screening asks which molecules may bind a given target; target fishing asks which targets may bind a given molecule.

Several *in silico* target-fishing strategies have been proposed, including ligand-based, receptor-based, pharmacophore-based, docking-based, and machine-learning approaches (Galati et al., 2021). PharmMapper is a representative example of reverse pharmacophore mapping. Given a query compound, it searches for good alignments between the compound and a database of pharmacophore models associated with protein targets (Liu et al., 2010). Its updated version expanded the target pharmacophore database and introduced statistically meaningful ranking scores for the identified targets (Wang et al., 2017). In simplified terms, PharmMapper asks whether the three-dimensional arrangement of chemical features in the query molecule can be fitted to pharmacophore patterns known or inferred for possible targets.

SwissTargetPrediction follows a different principle. It predicts probable targets of a small molecule by comparing the query compound to molecules with known bioactivity data. The method is based on the similarity principle: molecules that are similar in two-dimensional or three-dimensional chemical space are more likely to share biological targets (Daina et al., 2019). Thus, while PharmMapper is primarily organized around reverse matching against pharmacophore models, SwissTargetPrediction is organized around ligand similarity to compounds with experimentally known target annotations. Both tools are valuable references for BioSensIA because they address the same broad question: given a molecule, which biological targets should be considered plausible?

BioSensIA-DC differs from these tools in its representation-learning formulation. Instead of matching a query molecule against an explicit pharmacophore database, as in PharmMapper, or inferring targets from similarity to known ligands, as in SwissTargetPrediction, *BioSensIA-DC uses a learned compatibility model between molecules and protein pockets*. Its starting point is DrugCLIP, a contrastive learning model that represents molecules and protein pockets with separate encoders and trains them so that compatible molecule–pocket pairs are close in a shared latent space (Gao et al., 2023). DrugCLIP formulates virtual screening as a dense-retrieval problem: a pocket is encoded as a query vector, molecules are encoded as candidate vectors, and ranking is performed by comparing these vectors.

The encoders used by DrugCLIP are based on Uni-Mol, a three-dimensional molecular representation-learning framework that incorporates atom types and atomic coordinates (Zhou et al., 2023). A key feature of Uni-Mol is that it is designed to respect the geometry of molecular structures: its internal pair representations are built from interatomic distances, which are invariant to global translations and rotations of the molecule or pocket. At the same time, Uni-Mol includes an equivariant coordinate-prediction mechanism, so that when coordinates are predicted or updated, they transform consistently under rotations and translations of the input structure. In practical terms, this means that the representation is sensitive to the relative three-dimensional arrangement of atoms, but not to arbitrary choices of coordinate-system origin or orientation.

The central idea behind BioSensIA-DC is that the similarity score learned by DrugCLIP can be used in the reverse direction. If DrugCLIP can assign a compatibility score to a molecule–pocket pair for virtual screening, then the same score can be used to rank pockets for a fixed molecule. However, BioSensIA-DC is not merely a direct execution of the original DrugCLIP workflow. DrugCLIP provides the initial molecule–pocket representation framework, but the target-fishing formulation, candidate-pocket preparation, query construction, molecule-to-pocket retrieval pipeline, and initial diagnostic evaluation workflow were conceptualized and implemented from scratch as part of this work. The resulting prototype therefore adapts a virtual-screening representation model to the biosensor-oriented problem of prioritizing candidate bioreceptors for a query analyte.

1.3 Objectives and contributions

This paper describes BioSensIA-DC, a computational prototype for molecule-to-pocket target fishing in the context of bioreceptor discovery. The prototype extends the DrugCLIP representation framework beyond its original virtual-screening workflow. Instead of ranking molecules for a fixed protein pocket, BioSensIA-DC ranks candidate protein pockets for a fixed query molecule, producing a prioritized list of structural candidates for further investigation.

The implemented contributions include the construction of candidate-pocket files from DrugCLIP-compatible data, the construction of query-molecule files from SMILES strings, local DrugCLIP identifiers, PubChem CIDs, or compound names, and the implementation of molecule-to-pocket retrieval. The prototype also generates ranked pocket lists with compatibility scores, and provides an initial benchmarking and diagnostic workflow.

The present work should be read as a computational-prototype and methodology paper. Suitable benchmark validation has not yet been completed. Therefore, no claim is made that the current model is already a reliable target-fishing predictor. Instead, the paper reports the implemented workflow, the conceptual adaptation from virtual screening to target fishing, and the diagnostic issues that must be resolved before predictive performance can be assessed.